

## MINI PROJECT - LC09

### PARIS HOUSE PRICING PREDICTION MODEL USING BAYESIAN METHOD

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## 1. INTRODUCTION

The prices of houses in Paris, along with relevant information for each, are available in the dataset from <https://www.kaggle.com/datasets/mssmartypants/paris-housing-price-prediction>. The objective of this project is to create a prediction model for house prices in Paris using Bayesian methods. The dataset has 17 columns, and all attributes are numeric variables. The target variable here is price. The data contains the information's of size of a house in square meters, number of rooms in the house, does the house has a yard, does the house include a pool, how many floors that the house have, number of zip code, city's part range (the higher the value, the more exclusive the neighbourhood is), numbers of previous owners, the year that the house was made, is the house new or renovated, does the house have a storm protector, the house's basement size in meter, the house's attic size in meter, the house's garage size in meter, does the house have a storage room, does the house have a guest room, and the target variable, which is the house's price. The prediction will be done with the information that the dataset provided.

## 2. MODELS

To find the best model, we will standardise the features and use Gaussian Multiple Linear Regression as stated with the assumption of normality of the data. The alpha is the intercept when there isn't any features data. Beta is the intercept of each feature which then added all together.

$$Y_i = \alpha + \sum_{j=1}^{16} X_{ij}\beta_j$$

Model 1 : On the 1<sup>st</sup> model we will use Gaussian Multiple Linear Regression as stated. With  $X_{ij}$  being the value of each feature for each iteration of  $i$  with Priors  $\beta_j \sim Normal(0, 1000)$  and  $\sigma^2 \sim InvGamma(0.1, 0.1)$ .

Model 2 : On the 2<sup>nd</sup> model we will use Gaussian Multiple Linear Regression as stated but only using the features with strong correlations with the target feature based on frequentist linear regression model (**Image 1**) -> tolong ksh footnote (using features with 3 stars)

Model 3 : On the 3<sup>rd</sup> model we will use the 2<sup>nd</sup> Model Gaussian Multiple Linear Regression and features but with initial values from the frequentist linear regression model (**Image 1**) -> footnote 1 lagi.

### 3. COMPUTATION

All models will be fitted using the JAGS package in R. For building all of the Model the first 500 samples were discarded as burn in, then 2000 samples were drawn from each of the 3 chains with 0 thinning factor. On the 1<sup>st</sup> Model we are using all the dependent features of the data, but on the 2<sup>nd</sup> and 3<sup>rd</sup> Model we will only use seven selected features based on the frequentist linear regression model as shown on **image 1** and changing the initial values for the 3<sup>rd</sup> model with the result of the frequentist linear regression model. The model convergence is assessed by examining the trace plot, the Gelman-Rubin statistic (where a value < 1.1 indicates convergence), and the effective sample size (where a value > 1000 indicates convergence). The trace plots illustrate that all parameters in all models have achieved convergence. Gelman-Rubin statistics are computed for each parameter in every model, yielding multivariate potential scale reduction factors (PSRF) of 1.02 for the 1<sup>st</sup> Model, and 1.01 for 2<sup>nd</sup> and 3<sup>rd</sup> Model. Additionally, the effective sample size (ESS) for all parameters in all models is calculated, and in every model, ESS exceeds 1000. Therefore, we can conclude that all models have achieved convergence.

### 4. MODEL COMPARISON

The model performances are compared using the Deviance Information Criterion (DIC) and the Watanabe-Akaike Information Criterion (WAIC). The comparisons of the three models based on these criteria are summarized in Table 1. Among the four models, the one with the lowest DIC and WAIC is Model 3. Therefore, we will use Model 3 to predict house prices in this project.

**Table 1** Model comparisons

| Model          | DIC           | WAIC            | Multivariate PSRF |
|----------------|---------------|-----------------|-------------------|
| Model 1        | 339719        | 339719.8        | 1.02              |
| Model 2        | 339719        | 339719.9        | 1.01              |
| <b>Model 3</b> | <b>339719</b> | <b>323653.7</b> | <b>1.01</b>       |

### 5. RESULT

Applying Model 3, the posteriors of the parameters are summarized in Table 2. All parameters are significant, so we can conclude that in the real estate market of Paris, France :

- Increasing the land size (squareMeter) of the house by one multiplies its price by  $\exp(1.510)$ .
- Having a yard in house by one divides its price by  $\exp(0.616)$ .
- Having a pool in house by one divides its price by  $\exp(0.202)$ .
- Increasing the floors of the house by one divides its price by  $\exp(0.240)$ .
- Increasing the exclusivity of the house city part by one divides its price by  $\exp(1.403)$ .
- Newly built house by one divides its price by  $\exp(0.268)$ .
- Having a storm protector in house by one divides its price by  $\exp(0.492)$ .

Thus we can conclude that the price of a house mainly increased based on its land size.

**Table 2**

|          | Mean  | 2.5%   | 97.5% |
|----------|-------|--------|-------|
| $\alpha$ | 2.780 | -60.91 | 66.00 |

|                             |        |        |       |
|-----------------------------|--------|--------|-------|
| $\beta_{squareMeter}$       | 1.510  | -61.77 | 64.14 |
| $\beta_{hasYard}$           | -0.616 | -64.36 | 62.24 |
| $\beta_{hasPool}$           | -0.202 | -60.67 | 58.57 |
| $\beta_{Floors}$            | -0.240 | -60.41 | 60.75 |
| $\beta_{CityPartRange}$     | -1.403 | -62.46 | 60.47 |
| $\beta_{isNewBuilt}$        | -0.268 | -64.60 | 60.16 |
| $\beta_{hasStormProtector}$ | -0.492 | -61.83 | 64.34 |

## Lampiran

Image 1 :

```
Call:
lm(formula = price ~ squareMeters + hasYard + hasPool + floors +
    cityPartRange + isNewBuilt + hasStormProtector, data = df)

Residuals:
    Min       1Q   Median       3Q      Max
-6976.4 -1193.3    -5.9  1201.2  6968.3

Coefficients:
              Estimate Std. Error  t value Pr(>|t|)
(Intercept)  2.694e+02  7.279e+01  3.701e+00 0.000216 ***
squareMeters  1.000e+02  6.598e-04  1.516e+05  < 2e-16 ***
hasYard       3.012e+03  3.798e+01  7.930e+01  < 2e-16 ***
hasPool       2.978e+03  3.798e+01  7.841e+01  < 2e-16 ***
floors        5.457e+01  6.572e-01  8.303e+01  < 2e-16 ***
cityPartRange  4.717e+01  6.611e+00  7.135e+00  1.04e-12 ***
isNewBuilt    1.574e+02  3.797e+01  4.144e+00  3.43e-05 ***
hasStormProtector 1.421e+02  3.797e+01  3.742e+00  0.000183 ***
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Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1898 on 9992 degrees of freedom
Multiple R-squared:  1,    Adjusted R-squared:  1
F-statistic: 3.282e+09 on 7 and 9992 DF,  p-value: < 2.2e-16
```